

Support Vector Machine - Hard Margin

Perceptron and Margin

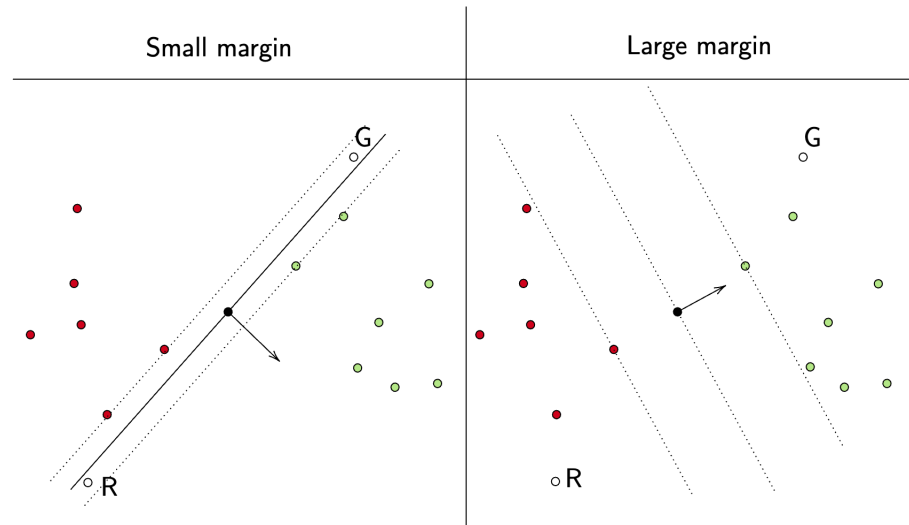
Radius Margin Bound

$$\text{number of mistakes} \leq \frac{R^2}{\gamma^2}$$

- $\|x_i\|^2 \leq R^2$
- $(w^{*T} x_i) y_i \geq \gamma; \gamma > 0$

Classifiers with a larger margin generalize better. That is, they perform better on test data-points compared to classifiers that have a small margin.

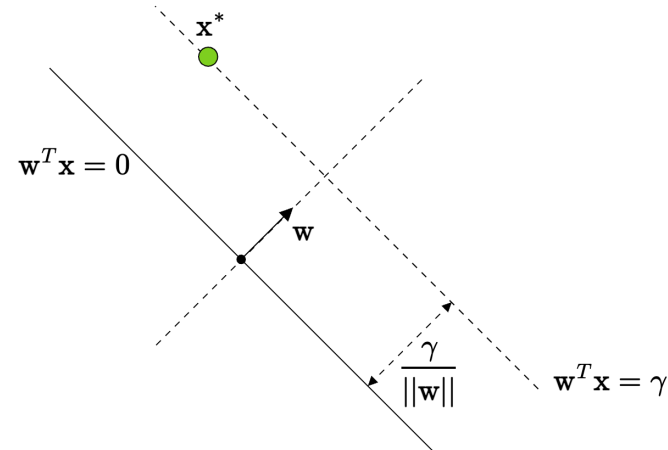
The number of mistakes made by the perceptron will depend upon the best possible w^* margin, that is larger $\gamma^2 \Rightarrow$ smaller proportion of mistakes.



Given a linear classifier, let us make the notion of margin more precise. There are two kinds of margin. Consider a linearly separable dataset with a positive margin. Let w be any valid classifier that perfectly separates the dataset. We can always find at least one point from the dataset that is closest to the decision boundary. Call this point x^* , For convenience, let this point have $y = 1$ This point will lie on some line parallel to the decision boundary, say $w^T x = \gamma$ This γ is termed the functional margin.

From our assumption, we have:

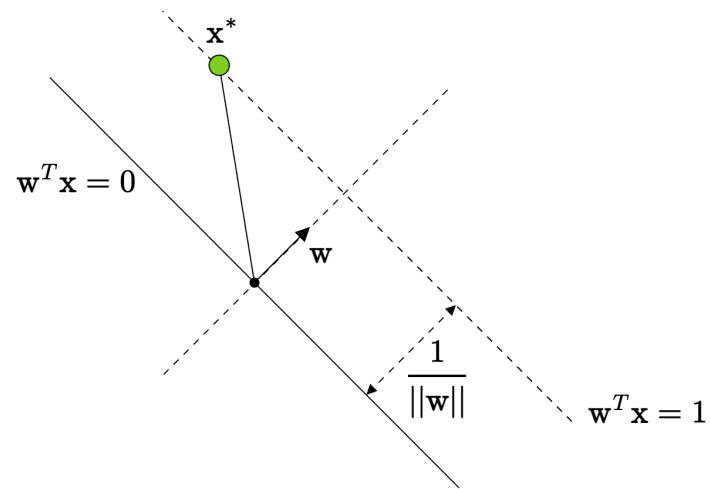
$$(w^T x_i) y_i \geq \gamma, 1 \leq i \leq n$$



The distance of between the lines $w^T x = 0$ and $w^T x = \gamma$ is called the geometric margin and is given by

$$\frac{w^T x^*}{\|w\|} = \frac{\gamma}{\|w\|}$$

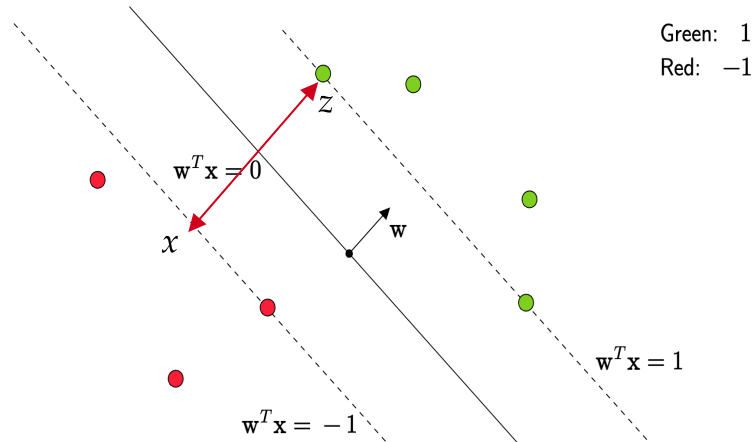
Since both w and γ can be scaled simultaneously without affecting the setup, we will set $\gamma = 1$



Therefore, for any linearly separable dataset with a positive margin, and a valid classifier w , we can set the functional margin $\gamma = 1$ and the geometric margin becomes

$$\text{Geometric Margin} = \frac{1}{\|w\|}$$

What is Width ?



$$\min \frac{1}{2} \|x - z\|^2$$
$$\text{s. t. } w^T x = +1$$

$$\text{width}(w) = \frac{2}{\|w\|^2}$$

$$\begin{aligned} \therefore \max_w & \frac{2}{\|w\|^2} \\ \text{s.t. } & (w^T x_i) y_i \geq 1 \end{aligned}$$

Equivalently,

$$\begin{aligned} \min_w & \frac{1}{2} \|w\|^2 \\ \text{s.t. } & (w^T x_i) y_i \geq 1 \quad \forall i \end{aligned}$$

Constraint Optimisation

$$\begin{aligned} \min_w f(w) \\ \text{s.t. } g(w) \leq 0 \end{aligned}$$

$$L(w, \alpha) = f(w) + \alpha g(w)$$

fix any w ,

$$\max_{\alpha \geq 0} L(w, \alpha) = \max_{\alpha \geq 0} f(w) + \alpha g(w)$$

$$\begin{aligned} \min_w f(w) \\ \text{s.t. } g(w) \leq 0 \end{aligned} \equiv \min_w \left[\max_{\alpha \geq 0} f(w) + \alpha g(w) \right]$$

If both function f and g are convex function, then we can swap min and max

$$\min_w \left[\max_{\alpha \geq 0} f(w) + \alpha g(w) \right] \equiv \max_{\alpha \geq 0} \left[\min_w f(w) + \alpha g(w) \right]$$

for multiple constraints,

$$\begin{aligned}
 & \min_w f(w) \\
 & \text{s.t. } g_i(w) \leq 0 \, \forall i, i \in 1 \dots k \\
 & \equiv \\
 & \min_w \left[\max_{\substack{\alpha_1 \dots \alpha_k \\ \geq 0}} f(w) + \alpha_1 g_1(w) + \dots + \alpha_k g_k(w) \right] \\
 & \equiv \\
 & \max_{\substack{\alpha_1 \dots \alpha_k \\ \geq 0}} \left[\min_w f(w) + \alpha_1 g_1(w) + \dots + \alpha_k g_k(w) \right]
 \end{aligned}$$

Solving the Optimization Problem

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem. We can recast it in the standard form as

$$\begin{aligned} \min_w \quad & \frac{1}{2} \|w\|^2 \\ \text{s. t.} \quad & (w^T x_i) y_i \geq 1 \quad \forall i = 1 \dots n \\ & \equiv \\ \min_w \quad & \frac{1}{2} \|w\|^2 \\ \text{s. t.} \quad & 1 - (w^T x_i) y_i \leq 0 \quad \forall i = 1 \dots n \end{aligned}$$

Objective Function

$$\begin{aligned} L(w, \alpha) &= \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i (1 - (w^T x_i) y_i) \\ \min_w \left[\max_{\alpha_k \geq 0} \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i (1 - (w^T x_i) y_i) \right] &\equiv \max_{\alpha_k \geq 0} \left[\min_w \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i (1 - (w^T x_i) y_i) \right] \end{aligned}$$

fix some $\alpha \geq 0$

$$\min_w \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i (1 - (w^T x_i) y_i)$$

Take gradient w.r.t w and set it to 0

$$w_{\alpha}^* + \sum_{i=1}^n \alpha_i (-x_i y_i) = 0$$

$$w_{\alpha}^* = \sum_{i=1}^n \alpha_i x_i y_i$$

In Matrix Notation,

$$w_{\alpha}^* = X Y \alpha$$

$$X = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} \quad Y = \begin{bmatrix} y_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & y_n \end{bmatrix} \quad \alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

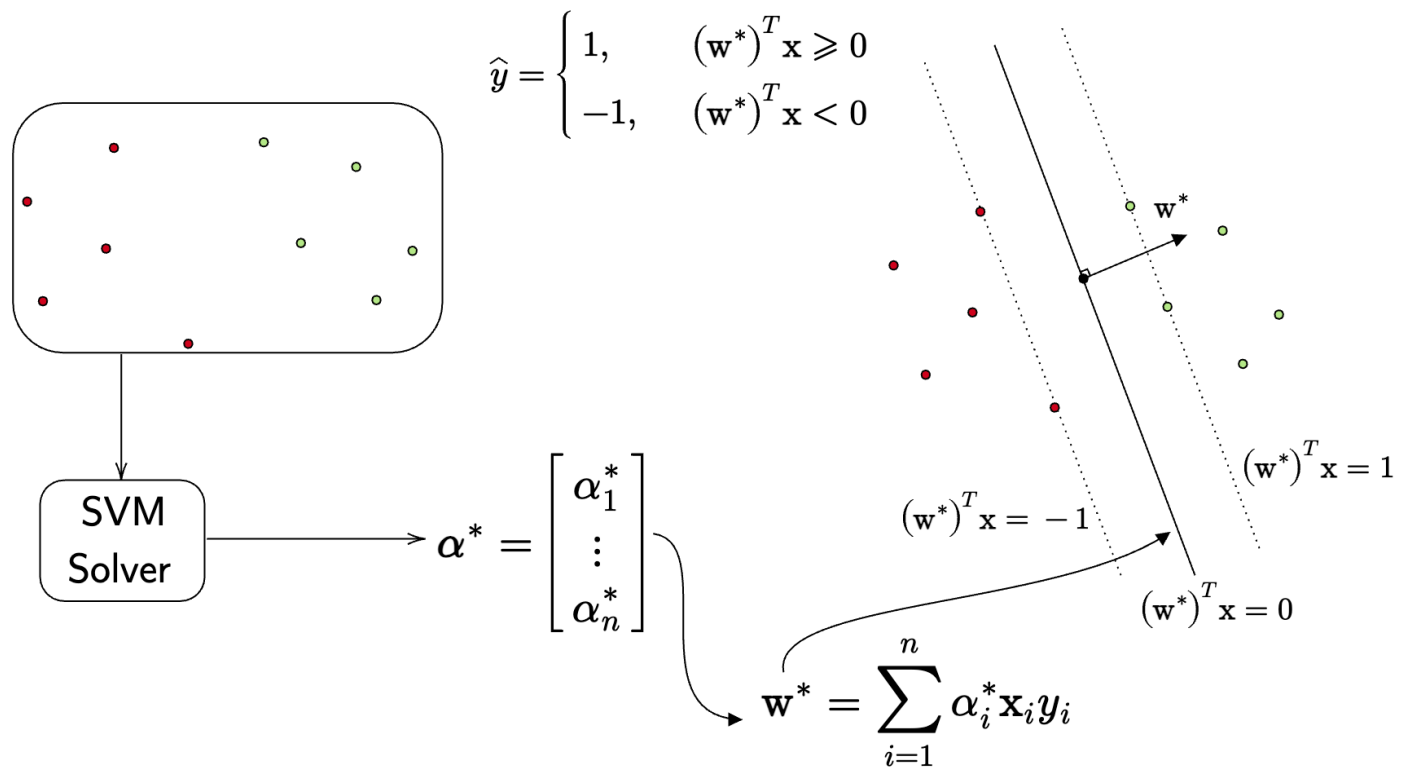
Plugging $XY\alpha$ back into the dual objective, we get the following form

$$\sum_{i=1}^n \alpha_i - \frac{1}{2}(XY\alpha)^T(XY\alpha)$$

The dual optimization problem therefore becomes:

$$\max_{\alpha \geq 0} \alpha^T \mathbf{1} - \frac{1}{2} \alpha^T Y^T X^T X Y \alpha$$

- Dual variable dimension in R^n while primal problem dimension is R^d
- Dual constraints are easier
- More Important, Dual depends on $x^T x$ and so can be kernelised

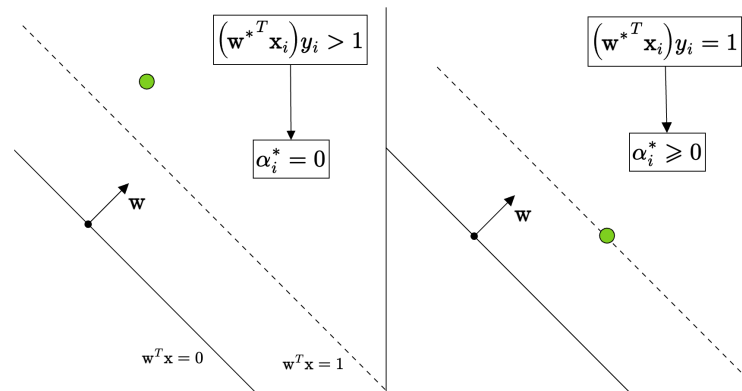


Complimentary Slackness

$$\alpha_i^* g_i(w^*) = 0 \forall i$$

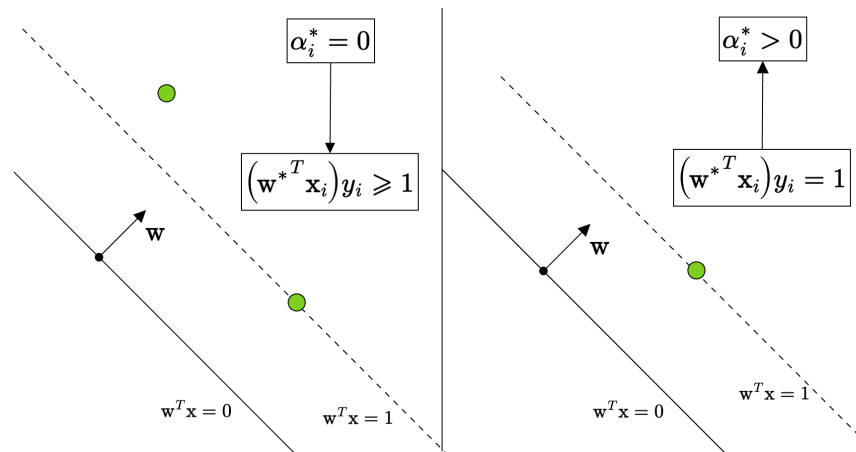
Primal View

- If $(w^T x_i) y_i > 1$, we have $1 - (w^{*T} x_i) y_i < 0$, which forces $\alpha_i^* = 0$. For a point which lies beyond the supporting hyperplanes, the value of α^* is zero.
- If $(w^T x_i) y_i = 1$, we have $1 - (w^{*T} x_i) y_i = 0$, which doesn't really forces α_i^* to be a particular value, So we just settle for $\alpha_i^* \geq 0$. For a point which lies on the supporting hyperplanes, we can't comment on the value of α^* but It can be any non-negative quantity.



Dual View

- If $\alpha_i^* = 0$, we can't comment on the position of the point. It could either lie on the supporting hyperplane or beyond.
- If $\alpha_i^* > 0$, we have $1 - (w^{*T} x_i) y_i = 0 \implies (w^{*T} x_i) y_i = 1$, A point with $\alpha_i^* > 0$ is going to lie on one of the supporting hyperplanes.



Some observations concerning support vectors

- All support vectors lie on the supporting hyperplanes.
- But note that every point on the supporting hyperplanes need not be a support vector.
- The weight vector can now be seen as a sparse linear combination of the data-points. The sparsity comes from the fact that all non-support vectors have $\alpha^* = 0$
- The support vectors are the most important training data-points as they lend their support in building the decision boundary.

Once a SVM has been trained, it can be used to predict the label of a test point like any other linear classifier

$$\hat{y} = \begin{cases} 1, & (\mathbf{w}^*)^T \mathbf{x}_{\text{test}} \geq 0 \\ -1, & (\mathbf{w}^*)^T \mathbf{x}_{\text{test}} < 0 \end{cases}$$

Linear SVM vs Kernel SVM

Linear SVM	Kernel SVM
$\mathbf{x}_i$	$\phi(\mathbf{x}_i)$
$\mathbf{X}$	$\phi(\mathbf{X})$
$\mathbf{X}^T \mathbf{X}$	$\mathbf{K}$
$\max_{\alpha \geq 0} \alpha^T \mathbf{1} - \frac{1}{2} \alpha^T (\mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}) \alpha$	$\max_{\alpha \geq 0} \alpha^T \mathbf{1} - \frac{1}{2} \alpha^T (\mathbf{Y}^T \mathbf{K} \mathbf{Y}) \alpha$
$\mathbf{w}^* = \mathbf{X} \mathbf{Y} \alpha^*$	$\mathbf{w}^* = \phi(\mathbf{X}) \mathbf{Y} \alpha^*$
$\hat{y} = \begin{cases} 1, & (\mathbf{w}^*)^T \mathbf{x}_{\text{test}} \geq 0 \\ -1, & \text{otherwise} \end{cases}$	$\hat{y} = \begin{cases} 1, & \sum_{i=1}^n \alpha_i^* k(\mathbf{x}_i, \mathbf{x}_{\text{test}}) \geq 0 \\ -1, & \text{otherwise} \end{cases}$

Reference

Arun Raj Kumar, Lecture Slides for Machine Learning Techniques
Karthik Thiagarajan, Machine Learning Techniques Github Notes.
Indian Institute of Technology Madras , CS2007 Course Website
Stanford University, CS229 Machine Learning Notes

Thank You !